Aether

# Data Analysis Report

Empirical diagnostic evaluation, descriptive statistics, and multivariate exploratory analysis.

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|---|---|
| DATASET NAME | **customer_metrics.csv** |
| REPORT DATE | 10 October 2026, 05:06 UTC |
| DATASET VERSION | Version 2 |
| TOTAL RECORDS | 20 rows |
| FEATURE DIMENSIONS | 4 columns (3 numeric, 1 categorical) |
| DATA QUALITY SCORE | **100/100** |
| SCOPE OF ANALYSIS | Full-Spectrum Exploratory & Statistical Audit |

**Notice:** This report is generated dynamically from verified local dataset records. All numerical metrics, distribution profiles, and correlation coefficients are computed strictly from the active dataset version.

## Executive Summary

| TOTAL RECORDS | TOTAL DIMENSIONS | DATA QUALITY SCORE |
|---|---|---|
| **20** | **4** | **100/100** |
| MISSING DATA RATE | DUPLICATE RECORDS | DETECTED OUTLIERS |
| **0.0%** | **0** | **0** |

This report presents a structured analysis for **customer_metrics.csv** (Version 2). The dataset comprises **20 observations** across **4 attributes**. Empirical evaluation yields an overall Data Quality Score of **100/100**, measuring record completeness (100%), uniqueness (100%), and validity (100%).

### Key Summary Findings:

• The dataset contains 20 verified observations across 4 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'tenure_months' and 'monthly_charges' (r = 0.97, strong positive).

• Categorical concentration observed in 'contract_type' where 'Two year' accounts for 60.0% of total entries.

## Dataset Overview

| DATASET NAME | **customer_metrics.csv** | ANALYSIS DATE | 10 October 2026, 05:06 UTC |
|---|---|---|---|
| ROW COUNT | 20 | COLUMN COUNT | 4 |
| FILE FORMAT | CSV | MEMORY FOOTPRINT | 1.7 KB |

### Column Overview:

| Column | Data Type | Non-null | Missing | Unique | Summary |
|---|---|---|---|---|---|
| **customer_id** | int64 | 20 | 0 (0.0%) | 20 | [1.0 to 20.0], Med: 10.5 |
| **tenure_months** | int64 | 20 | 0 (0.0%) | 20 | [1.0 to 63.0], Med: 26.5 |
| **monthly_charges** | float64 | 20 | 0 (0.0%) | 20 | [30.0 to 112.0], Med: 80.5 |
| **contract_type** | str | 20 | 0 (0.0%) | 3 | Top: Two year (60%) |

## Data Quality

| Quality Dimension | Score | Benchmark | Assessment Detail |
|---|---|---|---|
| **Completeness** | **100%** | >= 95.0% | Calculates proportion of non-null cells across the observation matrix. |
| **Uniqueness** | **100%** | >= 98.0% | Measures absence of identical duplicate row vectors. |
| **Validity** | **100%** | >= 90.0% | Evaluates presence of constant zero-variance features. |
| **Consistency** | **100%** | >= 85.0% | Checks for type homogeneity and uniform data formatting. |

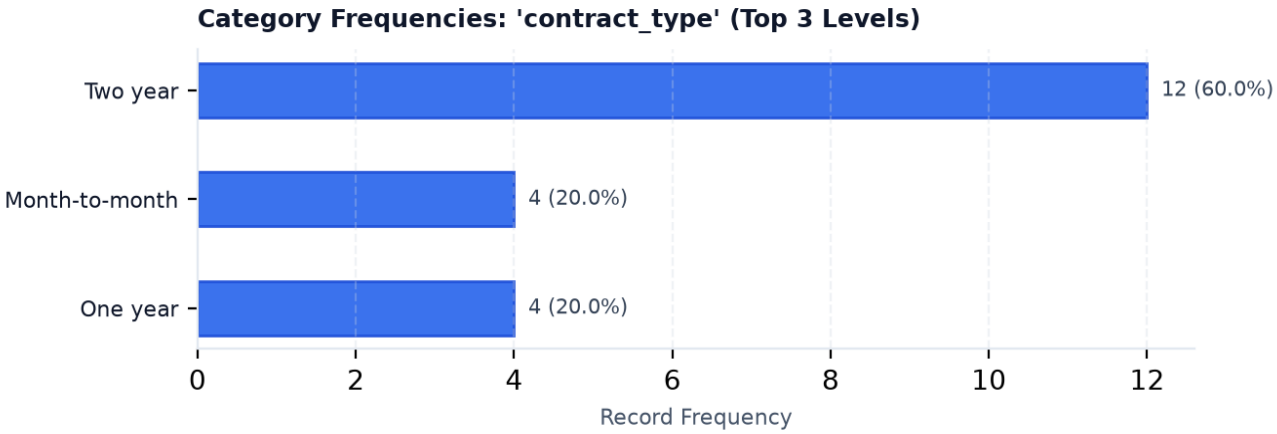
**Audit Details:** 0 missing cells identified (0.0% of dataset). Duplicate row analysis identified 0 duplicate record(s) (0.0%).

## Descriptive Statistics

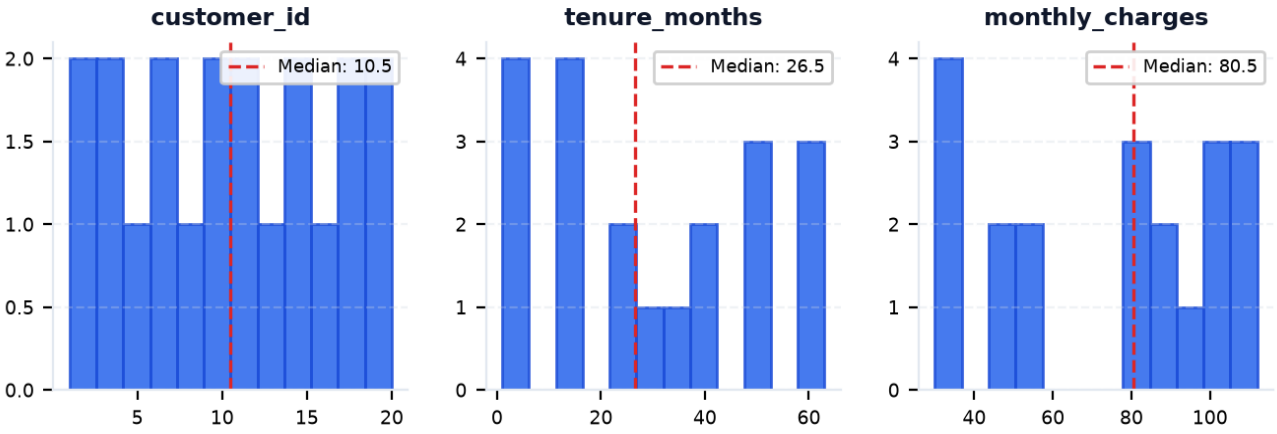
| Feature | Count | Mean | Median | Std Dev | Min | Max | Q1 | Q3 | IQR | Skew |
|---|---|---|---|---|---|---|---|---|---|---|
| **customer_id** | 20 | 10.5 | 10.5 | 5.9 | 1.0 | 20.0 | 5.8 | 15.2 | 9.5 | 0.00 |
| **tenure_months** | 20 | 29.6 | 26.5 | 21.1 | 1.0 | 63.0 | 13.5 | 48.5 | 35.0 | 0.20 |
| **monthly_charge** | 20 | 73.8 | 80.5 | 29.6 | 30.0 | 112.0 | 50.0 | 100.0 | 50.0 | -0.31 |

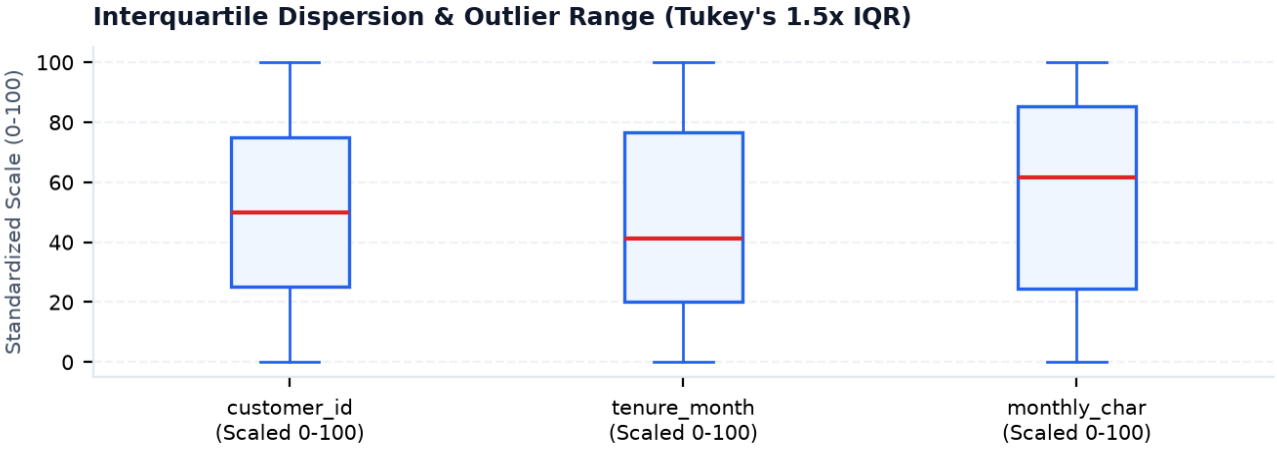
## Categorical Analysis

| Column | Unique | Most Frequent Value | Frequency | Percentage | Status |
|---|---|---|---|---|---|
| **contract_type** | 3 | Two year | 12 | 60.0% | Dominant (60.0%) |

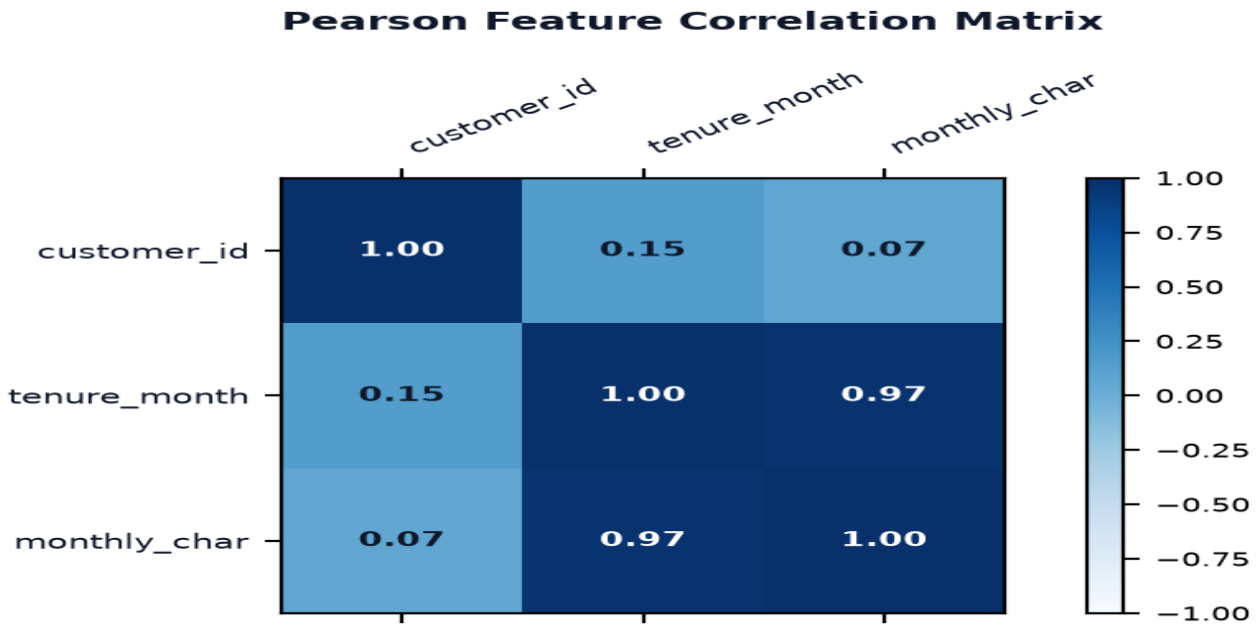



## Distribution Analysis



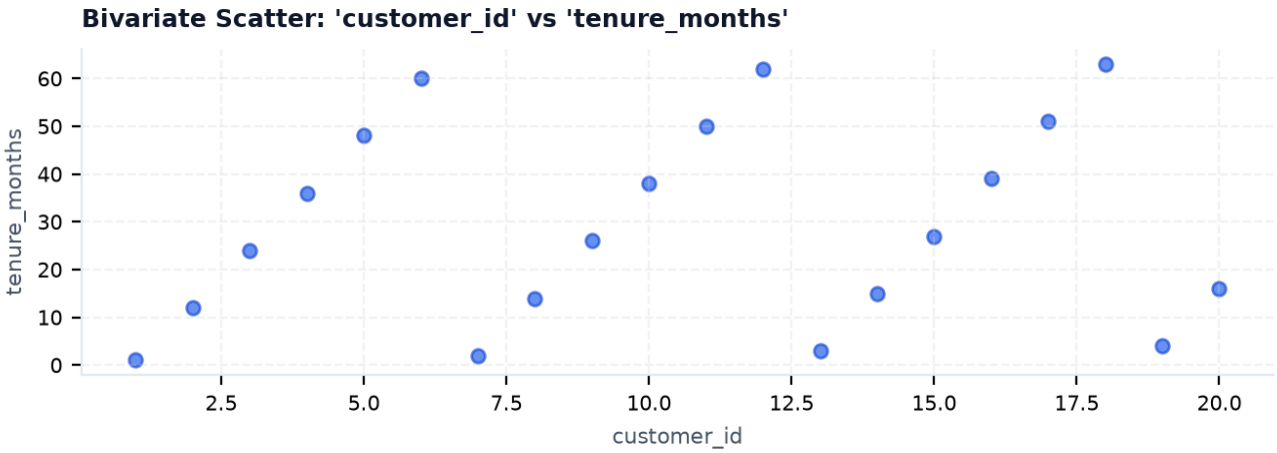



## Correlation Analysis




| Feature X | Feature Y | Pearson r | Strength | Direction |
|---|---|---|---|---|
| **tenure_months** | **monthly_charges** | 0.97 | Strong positive | Positive |
| **customer_id** | **tenure_months** | 0.14 | Little or no linear relationship | Positive |
| **customer_id** | **monthly_charges** | 0.07 | Little or no linear relationship | Positive |

## Relationships




## Outlier Analysis

| Feature | Outlier Count | Percentage | Lower Bound (1.5x IQR) | Upper Bound (1.5x IQR) |
|---|---|---|---|---|
| **customer_id** | 0 | 0.0% | -8.50 | 29.50 |
| **tenure_months** | 0 | 0.0% | -39.00 | 101.00 |
| **monthly_charges** | 0 | 0.0% | -24.99 | 174.98 |

No statistical outliers detected outside standard 1.5x IQR thresholds.

## Key Findings

• The dataset contains 20 verified observations across 4 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'tenure_months' and 'monthly_charges' (r = 0.97, strong positive).

• Categorical concentration observed in 'contract_type' where 'Two year' accounts for 60.0% of total entries.

## Recommendations

| Category | Priority | Action Recommendation |
|---|---|---|
| **Operational** | **Low** | Monitor the dominant category 'Two year' in 'contract_type' to identify potential sampling bias or operational dependency. Further domain validation is recommended. |
| **Analytical** | **Medium** | Evaluate 'tenure_months' and 'monthly_charges' for potential interaction terms or collinearity redundancy in multivariate regression. |
| **Data Quality** | **Low** | Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings. |

## Appendix

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|---|---|
| **Mathematical Methodology** | • **Tukey's IQR Outlier Rule:** Outliers identified strictly outside [Q1 - 1.5*IQR, Q3 + 1.5*IQR].<br>• **Pearson Correlation:** Linear association computed as r = cov(X, Y) / (std(X) * std(Y)).<br>• **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) measuring distribution symmetry. |
| **Data Quality Scoring** | Overall score computed via weighted evaluation: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale directly with missing cell frequency and duplicate rows. |
| **Compilation Provenance** | Generated by Aether Engine • ReportLab PDF Architecture • Python 3.0.6 / NumPy 2.5.3. Timestamp: 10 October 2026, 05:06 UTC. |